ASSIGNMENT 1

SOFTWARE ENGINEERING GROUP C

INDEX NUMBER: 4231230141

NAME: BENJAMIN BAIDOO

Q1a. FUNCTIONAL REQUIREMENTS

I. User Account Management: The system must allow users to register, log in, view/update their profile information, and reset passwords.
II. Book Search and Browsing: The system must allow users to search for books by title, author, ISBN, category, and browse books
III. Shopping Cart Management: The system must allow users to add books to a cart, view the cart contents, update quantities, and remove items.
IV. Order Placement: The system must allow authenticated users to initiate a checkout process, specify shipping details, and place an order.
V. Inventory Management The system must track the current stock level for each book and update the inventory upon a successful purchase.
VI. Payment Processing: The system must securely process payments for orders via integration with external payment gateways.
VII. Order History**:** The system must allow users to view their past orders and track the status of current orders.

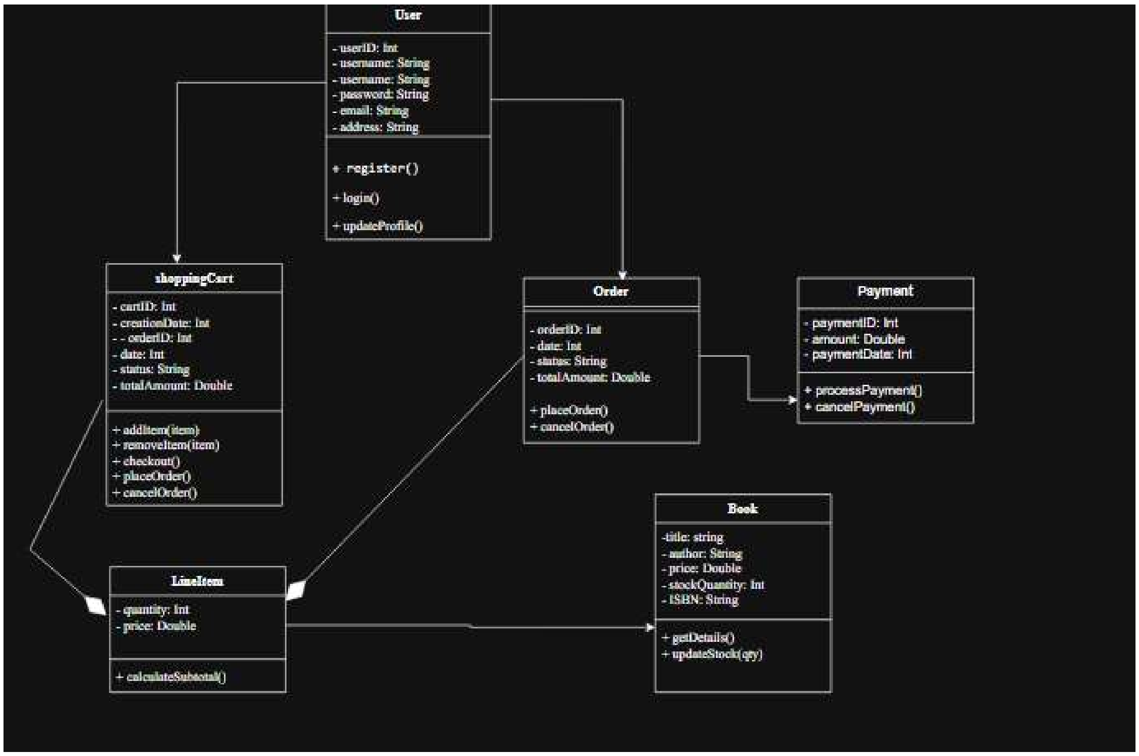
2. NON-FUNCTIONAL REQUIREMENTS

I. Security: The system must protect user account data and payment information using industry-standard encryption (e.g., SSL/TLS).
II. Performance: The book search and browsing features must return results within a maximum of 2 seconds, even with a large inventory.
III. Usability: The user interface must be intuitive, easy to navigate, and provide a consistent experience across different devices (responsive design)
IV. Scalability: The system must be able to handle a large number of concurrent users and a growing inventory without significant degradation in performance.

V. Reliability: The payment processing and order placement features must be highly reliable, with robust error handling and transaction logging.

Q1B

High-Level Class Diagram



Q1c. Modelling External Payment Gateway Interactions (10 Marks)

The interaction should be modeled as integration with an external service using secure API calls.

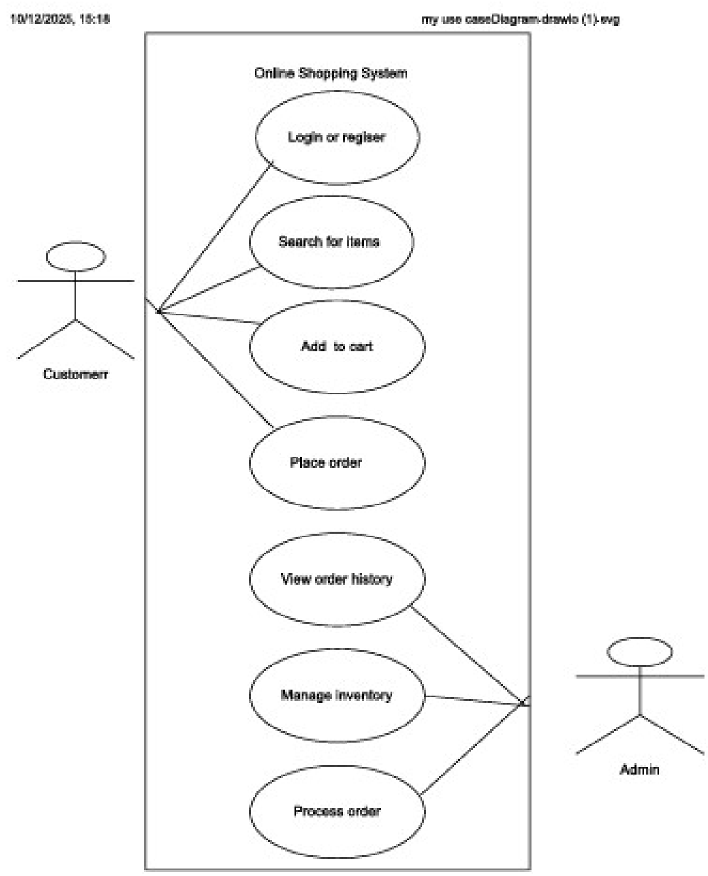
- Process Flow: The system's Order Processing Module acts as a client, sending payment requests (e.g., Authorize Payment, Capture Payment) to the External Payment Gateway service over HTTPS.
- Asynchronous Handling: The system must utilize Webhooks or IPNs (Instant Payment Notifications) from the gateway to handle final payment status updates asynchronously. The internal system tracks payment status (Pending, Authorized, Captured, Failed) via a state machine.

Security and Data Privacy Considerations

1. Tokenization: The e-commerce system should never store raw credit card numbers. Instead, the payment gateway issues a secure token representing the card, which is stored internally for transaction reference or future purchases.
2. Redirection/iFrame: The most secure method involves redirecting the user to a payment gateway-hosted page or using a secure iFrame. This ensures sensitive card data is entered and handled directly by the PCI-compliant gateway, not the bookstore server.
3. Encryption: All data transmission between the user's browser, the e-commerce system, and the payment gateway must use strong, up-to-date TLS/SSL encryption.

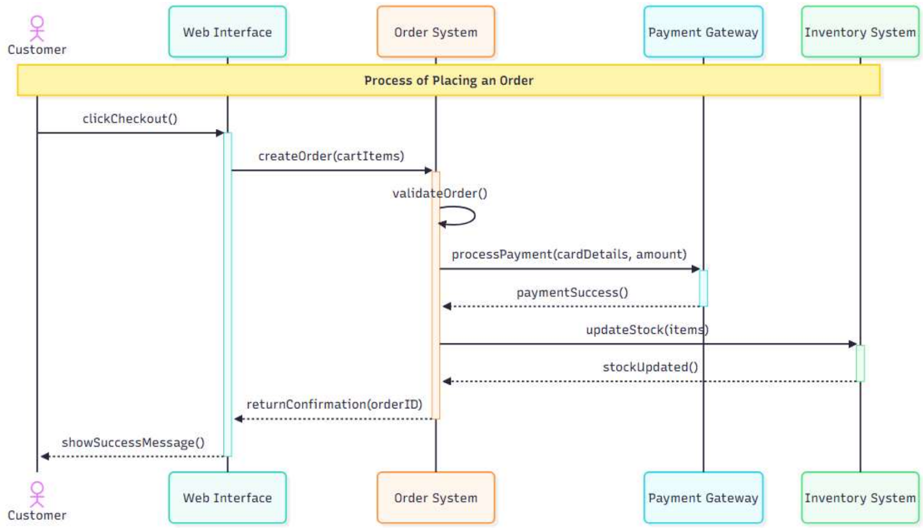
Q2a.

Use case diagram for an online shopping system



Q2b.

Sequence Diagram for Placing an Order



Q2c.

Significance of Deployment Diagrams in System Modelling

Deployment diagrams illustrate the physical topology of the system, showing where software components or artifacts are deployed on hardware nodes (servers, devices).

- It helps in Infrastructure Planning: Helps define the necessary server hardware, network connections, and middleware.
- Improves Performance and Scalability: Visualizes how components are distributed (e.g., application server vs. database server) to analyze traffic flow, performance bottlenecks, and plan scaling strategies.
- Security Architecture: Defines network boundaries and trust zones (e.g., separating the front-end web server in a DMZ from the internal database server).

Example of deployment diagram for a web application